

ComfyUI Task-to-Model Recommendation Guide

Hardware Optimization: RTX 5090 | Environment: Windows 11

This guide maps common image manipulation tasks to the specific models and support files found on your system to ensure maximum quality and speed.

Photorealistic Tasks

Task	Primary Model	Support Files (VAEs / CLIPs)	Recommendation & Strategy
Fixing Contrast & Brightness	Qwen Image Edit 2511	qwen_image_vae, t5xxl_fp16	Use for "Global Edits." Qwen is superior at understanding lighting instructions without changing the underlying image structure.
Adding Elements or People	FLUX.1 Fill (Dev)	ae.safetensors, t5xxl_fp16, clip_l	Use the 23.8GB full version. It provides the best environmental blending for new objects compared to the GGUF variants.
Removing Elements or People	FLUX.1 Fill (Dev)	ae.safetensors, t5xxl_fp16, clip_l	Better than generic inpainting. It "hallucinates" the missing background with higher fidelity, crucial for 4K workflows.
Combining Images (Blending)	DreamOmni2	flux1-kontext-dev, clip_l, t5xxl_fp16	Use the Vantage-DreamOmni2 node. It allows you to feed a source image and a reference image to transfer poses or objects seamlessly.
Restoring Old Photographs	FLUX.1 Kontext	ae.safetensors, t5xxl_fp16, clip_l	Best for "Structural Restoration." If the face is damaged, pair this with the InsightFace models (antelopev2) to preserve identity.

Creative & Non-Photorealistic Tasks

Task	Base Model	LoRA / Toolset	Strategy
Puppets / Claymation	FLUX.1 Dev	Master-Claymation.safetensors	Apply at a strength of 0.8–1.0. FLUX's prompt adherence ensures the "clay" texture applies to every detail.

Anime / Cartoons	FLUX.1 Dev	Flux Color Pop Sketch LoRA	This LoRA excels at turning photorealistic inputs into stylized, vibrant sketches or cell-shaded art.
Fun Stylized Changes	Z-Image Turbo	z_image_turbo_nvfp4	Use this for "Rapid Prototyping." Since it's optimized for your 5090, you can iterate through different fun styles in seconds.

💡 RTX 5090 Workflow Notes

1. The "No-GGUF" Policy for VRAM Giants

Since you have an RTX 5090, your VRAM is not a bottleneck.

- **Avoid:** Q8_0 or fp8 Text Encoders if a full version is available.
- **Always Use:** t5xxl_fp16.safetensors for your CLIP loader. This ensures the model understands complex prompts (like "holding a blue translucent orb") without the "brain fog" caused by quantization.

2. Identity Preservation (PuLID + InsightFace)

For any task involving people (Adding, Combining, or Restoring), use your **PuLID for FLUX** (pulid_flux_v0.9.1.safetensors).

- **Requirement:** Ensure your antelopev2 and buffalo_l ONNX models are loaded into the InsightFace node.
- **Benefit:** This prevents the AI from "genericizing" the face, keeping it looking like the specific person in your reference photo.

3. Background Removal (RMBG 2.0)

Before combining images or adding people, use **RMBG 2.0**. Removing the background of your source element *before* passing it to FLUX Fill or DreamOmni2 prevents "color bleeding" from the old background into your new scene.

4. Speed vs. Quality

- **For Speed:** Use z_image_turbo_nvfp4. It is built for your hardware and generates in ~1-3 seconds.
- **For Quality:** Use the **Full 35GB FLUX.2** or **FLUX Kontext**. These take longer (~10-15 seconds) but provide professional-grade skin textures and lighting.